

SAURABH SINGH

DOB: 10 JUNE 2001 Shahjahanpur, Uttar Pradesh

☎ (+91) 6386185878 ✉ saurabhchauhan6388@gmail.com in [saurabh](#) 🌐 [saurabh](#)

EDUCATION

National Institute of Technology, Rourkela

Master of Technology in Signal and Image Processing (CGPA - 7.83)

August 2025 – Present

Rourkela, Odisha

Axis Institute Of Technology and Management, Kanpur

B.Tech in CSE(Artificial Intelligence and Machine Learning)(CGPA - 7.23)

2020 – 2024

Kanpur, Uttar Pradesh

COURSEWORK / SKILLS

- Probability
- Statistics
- Data Analysis
- Data Science
- Python Programming
- Artificial Intelligence
- Machine Learning
- DBMS

PROJECTS

Smart Traffic Monitoring System 🗄 | [Streamlit](#), [Python](#), [Deep Learning](#)

2025

- Developed an advanced Smart Traffic Monitoring System to detect, track, and analyze vehicles in real-time using deep learning models such as YOLOv8 and DeepSORT, improving traffic surveillance and management efficiency.
- Implemented techniques including object detection, multi-object tracking, and bounding box localization to process video streams and accurately identify vehicles with unique IDs.
- Designed a scalable pipeline for traffic analysis by integrating vehicle counting and real-time video processing, enabling insights into traffic density and flow for smart city applications.

Image Search Engine 🗄 | [Streamlit](#), [Python](#), [Image Processing](#), [Machine Learning](#), [Deep Learning](#) **2024**

- Developed an innovative Image Search Engine project aimed at efficient visual search capabilities. It provides 5 most similar results of a query image.
- This Image Search Engine project offers a user-friendly interface, allowing users to upload images to initiate the search process.
- Leveraging the power of convolutional neural networks (CNNs) and transfer learning, the system provides accurate and efficient image retrieval.

TECHNICAL SKILLS

Languages: C, Python, SQL

Technologies/Frameworks: Scikit-learn, Keras, PyTorch, Pandas, CrewAI

Developer Tools: VS Code, PyCharm, Jupyter Notebook

CERTIFICATIONS

- Crash Course on Python - Coursera
- Learning Data Analytics - LinkedIn Learning
- Learning Data Visualization - LinkedIn Learning

Extracurricular Activities

Debate Team

Team Member

07-2021 – 05-2023

Axis Colleges

- Participated in inter-university debate competitions.
- Developed skills in public speaking, critical thinking and argumentation.